Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/07/2024
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	11/04/2021	**End Date:**	04/30/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/30/2024 per NSSC information (Ed., 8/31/24) NOTE: End date changed to 11/03/2023 per NSSC information (Ed., 11/3/22)		
Key Personnel Changes/Previous PI:	This is a transfer of funding from Dr. Elison Blancaflor (formerly at the Noble Research institute) to the University of Wisconsin-Madison. The postdoc working on the project (Dr. Sabrina Chin) moved from the Noble Research Institute to the University of Wisconsin-Madison to continue with the work and is now a CoI on the project.		
COI Name (Institution):	Chin, Sabrina Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	80NSSC22K0029		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: This project is a continuation with Principal Investigator (PI) Dr. Simon Gilroy at the University of Wisconsin of "The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth," grant 80NSSC18K1462. The previous PI was Dr. Elison Blancaflor, Noble Research Institute, from July 2018-March 2021. Plants will be a major component of advanced life support systems to enable NASA to realize its vision of long duration space exploration that will take humans to Mars and beyond. However, plants can only be effectively used for such systems if we have a deeper mechanistic understanding about how their development is regulated by the spaceflight environment. Recent "omics" studies have shown that plants reprogram their genetic circuitry when grown in the near weightless conditions of space. Genes involved in cell wall remodeling and oxidative stress are major gene categories that are differentially regulated by microgravity. Further, it has been shown that microgravity in space and simulated microgravity using clinostats or random positioning machines elicit roughly similar root skewing responses in the model plant Arabidopsis thaliana. Root skewing was more pronounced in mutants to a root-expressed vegetative actin isoform. Consistent with the enhanced root skewing of actin mutants in microgravity is the observation that other plant species such as maize (Zea mays L.) have stronger root directional growth responses on a clinostat when chemicals such as latrunculin B disrupt their actin cytoskeleton. Based on these observations, we will test the hypothesis that the actin cytoskeleton is a key player in dictating root growth directionality in microgravity and simulated microgravity by regulating the process of autotropic organ straightening. Moreover, we hypothesize that a set of cell wall cross-linking (i.e., class III peroxidases) and putative actin-regulatory proteins (i.e., HLB1 and MIN7/BEN1) mediate root hair elongation and directional root growth, respectively, under microgravity and simulated microgravity. Our hypotheses will be addressed through three specific aims including the analysis of cell wall composition/distribution in wild-type Arabidopsis roots, and detailed characterization of selected Arabidopsis cell wall and actin mutants subjected to microgravity simulation on a slow rotating clinostat and random position machine (aim 1 and 3). We will also use microgravity simulators to address the hypothesis that the plant hormone brassinolide regulates directional root growth in maize through its modulation of actin organization and dynamics (aim 2). Our research plan builds on successes of previous NASA-funded ground and spaceflight results, and outlines a comprehensive set of experiments including cell wall immunocytochemistry, computer-based root growth analyses, actin quantification, and genetics. The proposed work aligns with Plant Biology (PB) Element of the Space Biology Science Plan. Results obtained from plant experiments on microgravity simulators will address guiding questions linked to the decadal survey recommendations including the impact of gravity on plant growth, development and metabolism (PB-1), and molecular mechanisms underlying how plants sense and react to gravity (PB-5). We expect that the proposed research will provide us with basic knowledge that will guide the development of plant cultivars better adapted to the microgravity environment of spaceflight, an outcome that will prove beneficial for the design of plant habitats on future space colonies and for addressing great agricultural challenges on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to gain a better mechanistic understanding of how plant roots function at the cellular and molecular levels. A primary focus of the proposed work is to ask how the microgravity environment modulates the directionality of root growth. Because roots are essential for the uptake of water and nutrients, knowledge gained from successful implementation of the proposed work will be crucial in guiding pre-breeding efforts aimed toward the development of crops with improved productivity through better and more efficient root systems. Thus, agriculture on Earth is one beneficial outcome of this project.
	OVERVIEW: This research has used microgravity analogs to test hypotheses about plant responses to reduced gravity generated from the previous plant-focused spaceflight research campaigns of Biological Research in Canisters-16 (BRIC-16) and Advanced Plant Experiments-03 (APEX-03-1). BRIC-16 and APEX-03-1 revealed that microgravity had a profound effect on root growth directionality, cell wall structure and composition, and the expression of a suite of root-specific genes involved in regulating cell wall function. Moreover, this research implicated the actin cytoskeleton as a major player in modulating root growth in microgravity. These observations led to the overarching hypothesis tested in this current research project that microgravity modifies root growth directionality in large part by modifying the expression of genes that link the actin cytoskeleton to structural remodeling of the cell wall. To test this hypothesis the following specific aims were pursued: Specific Aim 1. Immunocytochemistry of selected cell wall components in roots of Arabidopsis thaliana (A. thaliana) seedlings grown on the clinostat to simulate a microgravity environment. Specific Aim 2. Characterizing autotropic root straightening in maize roots in response to brassinolide (BL) - and latrunculin B (LatB) - induced actin disruption. Specific Aim 3. Evaluation of root growth of selected Arabidopsis thaliana actin/cell wall-related mutants grown in simulated microgravity generated through clinorotation. RESEARCH ACCOMPLISHMENTS: SPECIFIC AIM 1. Immunocytochemistry of selected cell wall components in roots of Arabidopsis thaliana seedlings grown on a slowly rotating clinostat. Status: completed. The APEX-03 experiment grew plants on the International Space Station for either 6 or 11 days. These samples were returned for glycomic analysis of cell wall polymer composition. Glycomics uses enzyme-linked immunosorbent assay (ELISA) to quantify non-cellulosic cell wall polymer levels via a battery of monoclonal antibodies raised against specific cell wall components. APEX-03 also returned fixed materials allowing for confirmation of the ELISA results using immunocytochemistry. Xyloglucans are the most common hemicellulose in dicot cell walls and the glycomics provided a survey-level monitoring of wall composition that highlighted xyloglucans as likely being modified in spaceflight. To validate these results, 22 of the monoclonal antibodies that recognized a range of wall polymers including xyloglucan, xylan, and arabinogalactan epitopes were chosen from the glycomics analysis. These antibodies were then used for immunocytochemistry to localize where these polymers were likely to be changing. These analyses revealed the spatial patterning underlying the spaceflight-triggered changes suggested by the glycomics survey. For example, the

Task Progress:

XG-4 non-fucosylated-xyloglucan recognized by antibody M58 was found to be most intensely deposited in the periphery of the spaceflight-grown roots, whereas the xylan backbone epitope recognized by antibody M140 was more intensely labeled in the xylem of space-grown roots. These results suggest that the spaceflight environment triggers changes in xyloglucan-related polymer production, where distinct spatial patterning of their production is an important element of the response (Nakashima et al., 2023).

The samples used for glycomics were also used to extract RNA for parallel transcriptomic profiling. Analysis of this APEX-03 RNAseq data was made using the iDEP2.1 platform with a false discovery rate set to <0.05 and a fold-change cutoff of 1.5x. This study revealed distinctly different spaceflight-responsive transcriptome profiles of root versus shoot tissues that changed in the plants sampled at day 11 compared to those harvested at day 6. Gene ontology enrichment analysis highlighted cell walls as a likely important element of these transcriptional responses

The glycomics results from aim 1 showed change in xyloglucan-related wall polymers is triggered by growth in spaceflight. However, analysis of the expression patterns of the 13 genes that form the xyloglucan biosynthetic pathway in Arabidopsis from the APEX-03 RNAseq data do not support the hypothesis that transcriptional regulation of this pathway explains these changes in cell wall biochemistry. It is likely that spaceflight is altering cell wall structure through effects on post-transcriptional regulatory events.

SPECIFIC AIM 2. Characterize autotropic root straightening in maize roots in response to brassinolide (BL)- and latrunculin B (LatB)-induced actin disruption. Status: completed at the Noble Research Institute.

This research revealed that the plant hormone, epi-Brassinolide (eBL) is involved in autotropic root straightening post gravistimulation through its effects on F-actin in roots of maize (Zea mays). Treatment with eBL resulted in the over-shooting of the bending of roots after a gravity stimulus was removed by placing the plants on the clinostat. This effect was caused by clinorotation-linked disruption of F-actin organization and dynamics. This outcome was published in the journal, Frontiers in Plant Science (de Bang et al., 2020). This work has also led to the development of live cell actin reporters in M. truncatula. Plants were genetically engineered to stably express Lifeact-mGFP and GFP-ABD2-GFP that produced high quality labeling of the actin cytoskeleton in roots and leaves with minimal disruption in overall growth. These transgenic lines were then used to study the effect of eBL on F-actin in M. truncatula root cells (de Bang et al., 2020). Additionally, methods have been developed for live cell imaging of F-actin reporters in A. thaliana and M. truncatula that were published as a chapter in the book, Cytoskeleton Methods and Protocol (Chai et al., 2022). [Ed. Note: For complete citations, see References below and Cumulative Bibliographies under PI Gilroy (Grant #80NSSC22K0029) and former PI Blancaflor (Grant #80NSSC18K1462).]

SPECIFIC AIM 3. Evaluate root growth of selected Arabidopsis thaliana actin/cell wall-related mutants on microgravity simulators. Status: completed.

Three recessive, non-allelic actin-related mutants have been isolated from a T-DNA mutagenized population of A. thaliana in a forward genetic screen for increased sensitivity of root growth inhibition to LatB. This research avenue started in previous NASA-funded research, Gramt #NNX12AM94G, to find novel players in actin-regulated root growth by searching for LatB hypersensitivity. These mutants were named hypersensitive to LatB (hlb) 1, 2, and 3. The hlb1 mutation was mapped to a gene of unknown function that encoded a protein with a six tetratricopeptide repeat motif structure during the previous funding. Analyses characterized it to have functions in modulation of actin dynamics at the trans-Golgi network/ early endosome, resulting in a publication in The Plant Cell (Sparks et al., 2016). Continuing this work, the hlb3 mutation has now been mapped to exon 2 of the gene, FORMIN HOMOLOGY 20 that is involved in the regulation of actin cytoskeleton. This work involved using a new tool, TDNAscan, that demonstrated improved accuracy in the identification of complete and truncated T-DNA insertions, and was published in Frontiers in Genetics (Sun et al., 2019). The hlb2 mutation has now also been mapped to the gene, SPIRRIG (SPI). Characterization of its function showed it played a key role in the regulation of root hair elongation by maintaining F-actin dynamics at the site of active root hair growth (Chin et al., 2021).

In parallel studies to the LatB hypersensitive mutants, analyses have been performed on the effect of growth in simulated microgravity on root growth and orientation of actin mutants: act2-3, act7-4, act8-4, act2-3 act8-2, along with knockout mutants in SPI. Simulated microgravity was generated using custom-built 2-D clinostats. Previous research has shown that mutants in ACT2 consistently showed strong root skewing and coiling responses in spaceflight during BRIC-16 (Nakashima et al., 2014). This current analysis has further revealed that ACT2 (and potentially SPI) as a key player, required for the inhibition of root growth and enhanced skewing triggered by growth in simulated microgravity on the clinostat.

IN SUMMARY, the major findings from this research have been: (1) Spaceflight is associated with a remodeling of the plant cell wall with major effects on xyloglucan polymer levels likely controlled at the post-transcriptional level; (2) The actin cytoskeleton is a major driver of root elongation and gravity-driven directional growth, with key roles for actin isoform 2 and the SPIRRIG protein; (3) Root straightening after gravstimulation is regulated through brassinosteroid and actin-dependent processes.

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